

Enterprise AWS modernization

Story A

Separate environments.

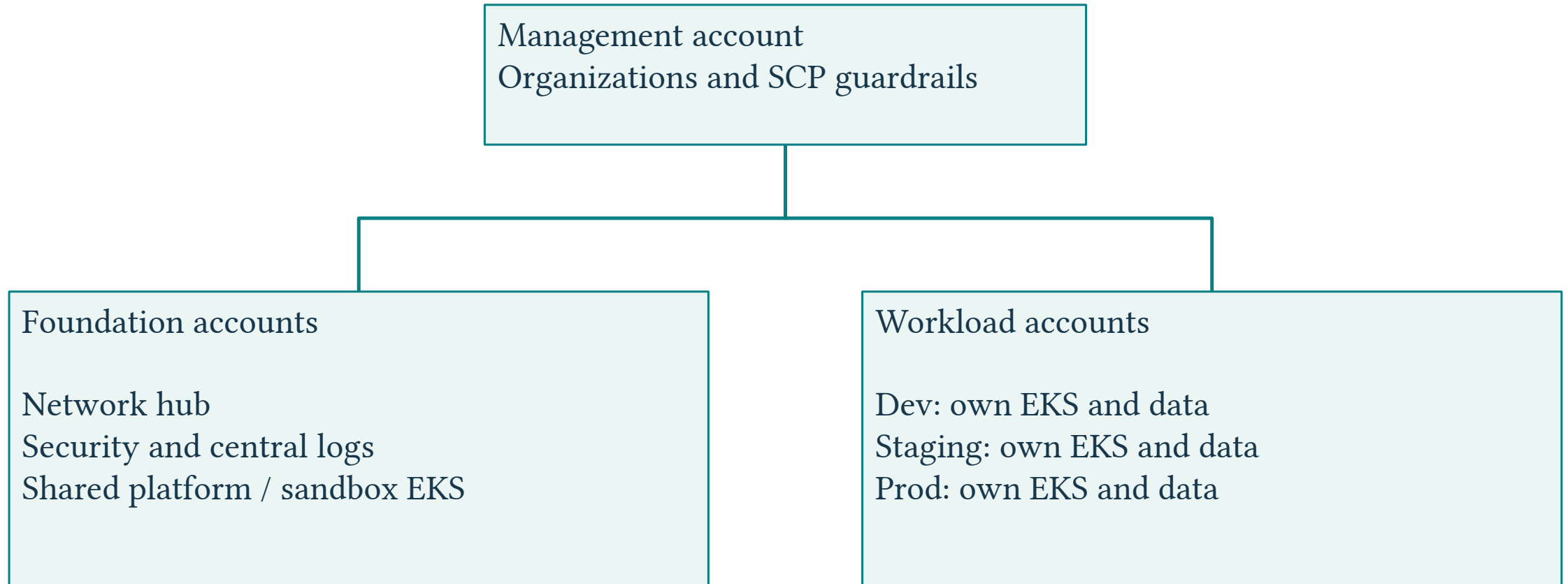
Repeatable releases.

Faster diagnosis.

Hussen Ali

Architecture proposal and delivery plan

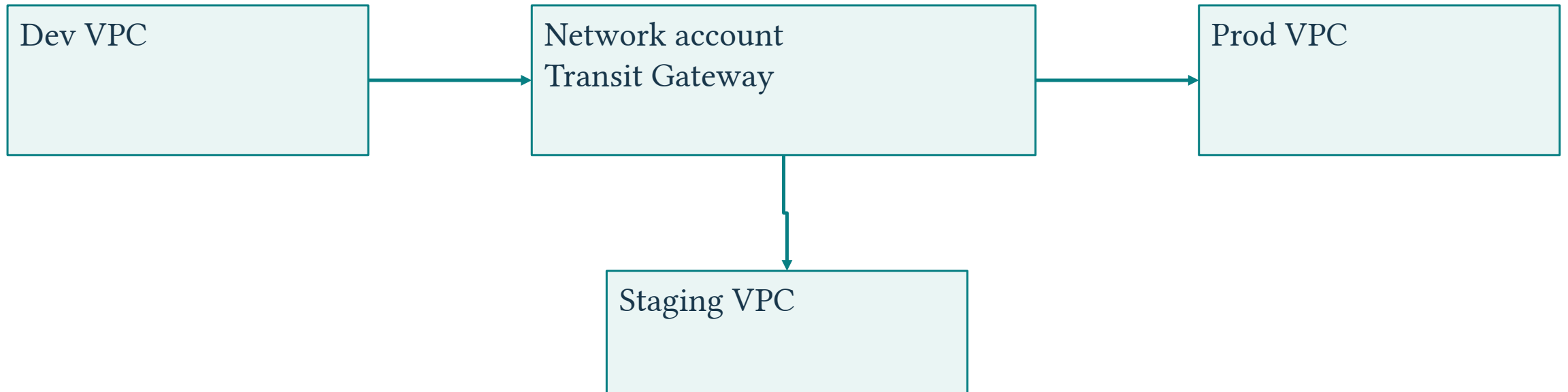
Multi-account structure



Separate account permissions and state for every environment

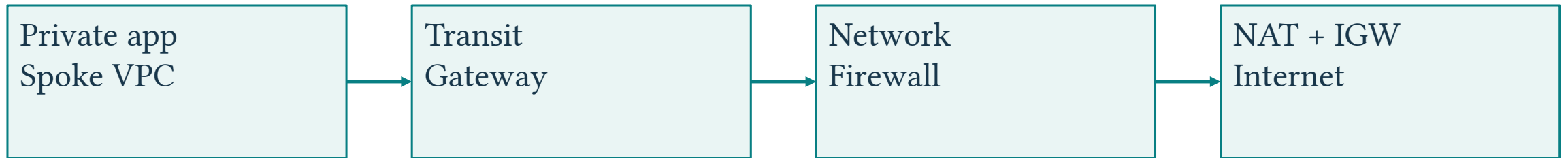
Hub-spoke network topology

Approved paths only; separate TGW route tables



Private app subnets and isolated data subnets in each spoke

Central internet egress



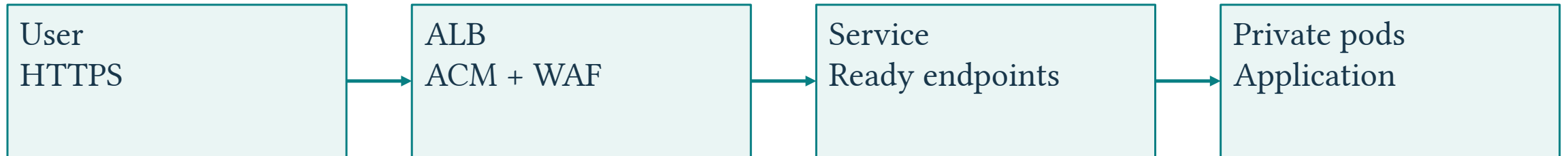
Same inspection path for the return traffic

One firewall/NAT path per AZ, with tested failure behavior

Data subnets: no direct internet default route

EKS request and authentication flow

DNS: Route 53 User login: application / OIDC

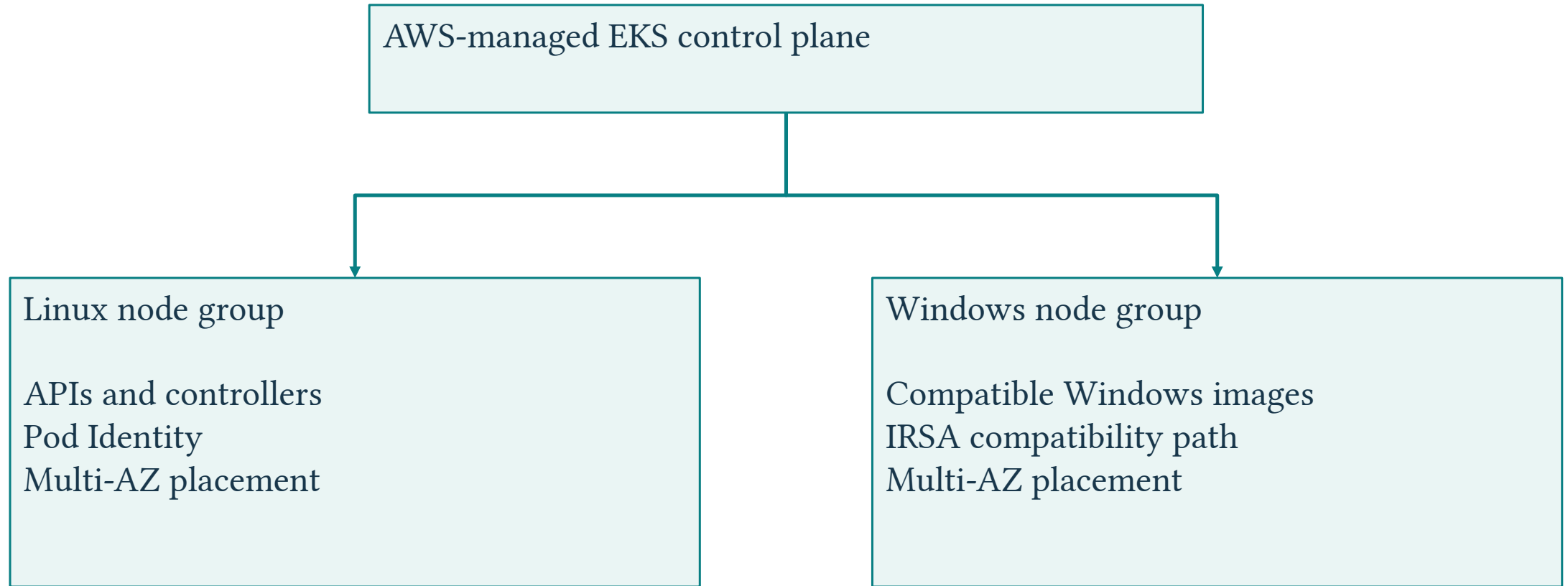


Pod access to AWS: service account and scoped IAM role

Administrator access: private EKS API, access entries and RBAC

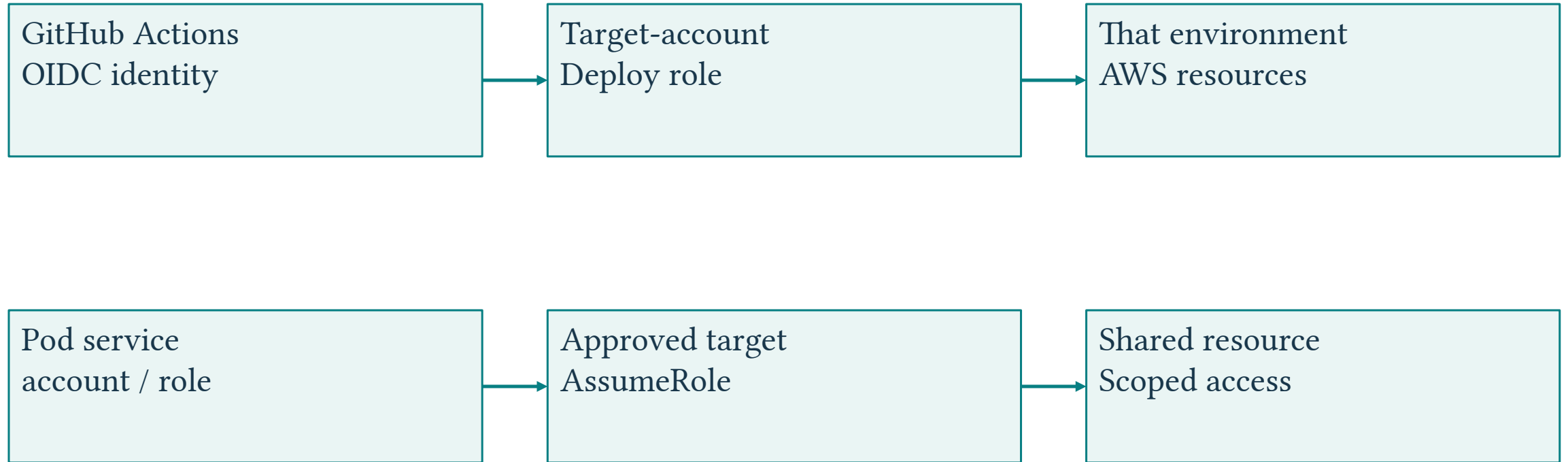
Ingress accepts requests. Workload IAM controls AWS API access.

Linux and Windows on EKS



OS labels control placement. Windows CNI limits need explicit review.

Cross-account IAM model



SCPs and permission boundaries limit access; IAM policies grant it

Separate Terraform repositories

Reusable modules repository

modules/
vpc/ tgw/ firewall/
eks/ iam/ observability/

Each module: main.tf, variables.tf, outputs.tf,
versions.tf

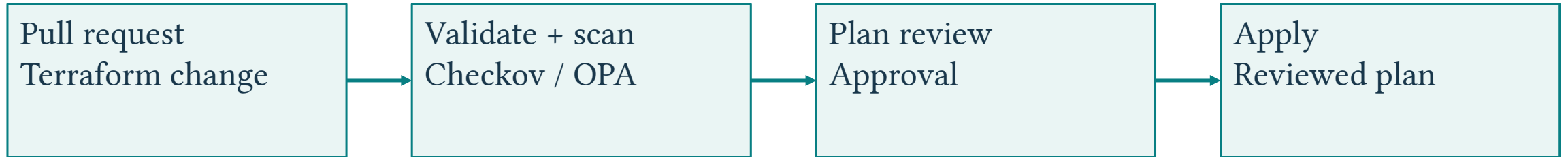
Account / foundation repository

management/ network/ security/
platform/
environments/dev/
environments/staging/
environments/prod/
backend/ policies/

Versioned module calls flow into every account root

Application repository is separate: charts/ values/ argocd/

Terraform change and state flow

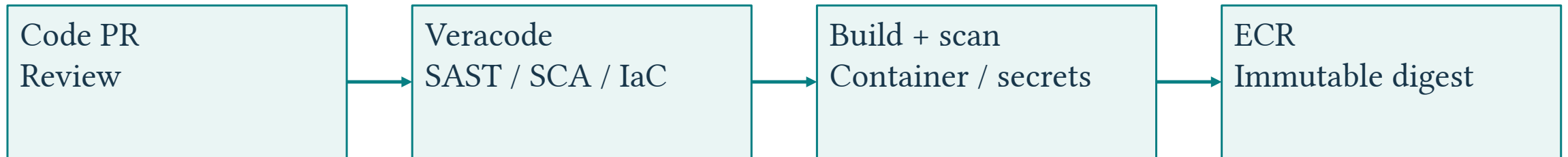


Separate encrypted, versioned S3 state per account and stack

Assignment baseline: DynamoDB lock; agree native S3 locking

Dev approval cannot authorize a production apply

Application CI: Veracode and release gates

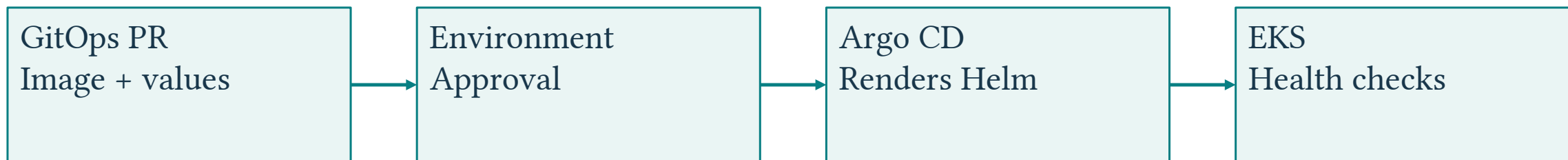


Policy failure blocks publishing or promotion

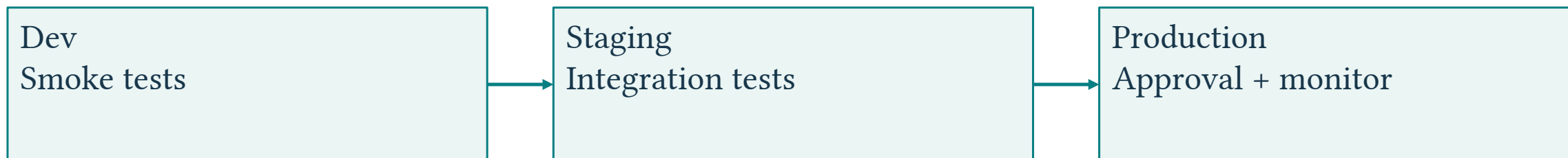
CI output: tested image, scan evidence and GitOps change request

Use the client-approved scanner. Do not claim every Veracode capability is enabled until confirmed.

Application CD: Helm and Argo CD



Promote the same image digest



Rollback: revert the Git version, sync, verify recovery

Observability and centralized evidence

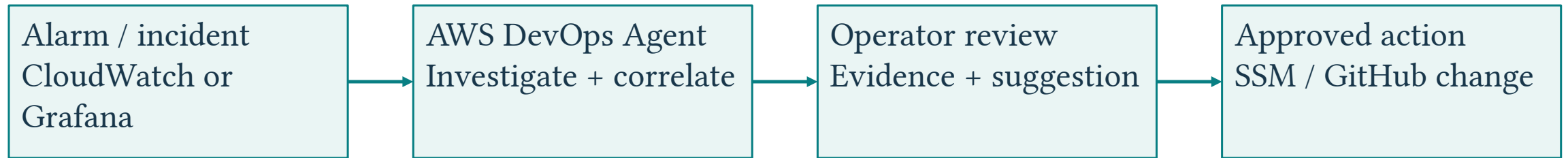


Kubernetes metrics: Prometheus and Grafana

Optional extension: application tracing after a clear need and pilot

Common incident ID links logs, metrics and diagnostic evidence

Incident response with AWS DevOps Agent



Agent suggests. The on-call owner approves or rejects the change.

Keep a manual runbook path when the agent is unavailable

Your POC is a credible optional enhancement, not a claim of full production rollout.

Hybrid OS operations

Linux checks	Windows checks
Service state and journal	Services and Event Viewer
Memory, disk and inodes	Memory, disk and IIS app pools
Nginx / Apache / container logs	IIS / application / container logs
Bash via approved SSM document	PowerShell via approved SSM document

Patch EC2 in maintenance windows. Replace EKS nodes in safe batches.

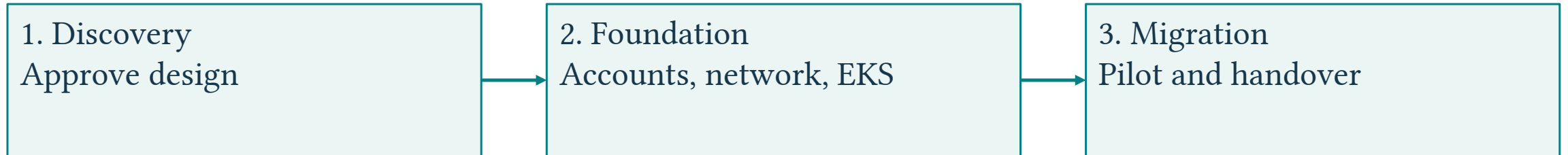
Well-Architected controls

Pillar	Concrete control
Operational Excellence	GitOps releases and tested incident runbooks
Security	Account isolation, scoped IAM and inspected egress
Reliability	Multi-AZ capacity and tested recovery
Performance Efficiency	Resource requests, load tests and autoscaling
Cost Optimization	Right-sizing, ownership tags and log retention
Sustainability	Remove idle capacity and expire unneeded data

Key decisions and trade-offs

Decision	Benefit	Trade-off
Cluster per environment	Stronger isolation	More platform cost
Central egress inspection	Consistent network policy	Shared dependency and routing
Managed Linux/Windows groups	Common deployment platform	Windows compatibility limits
Helm + Argo CD	Reviewable desired state	Team onboarding and rollback rules

Delivery plan and acceptance



Acceptance: isolation tests, successful releases and recovery game day

Target: common-incident MTTR under 15 minutes, measured in tests

Weekly client review, named owners and written change approval

Appendix A: TGW attachment sample

```
resource "aws_ec2_transit_gateway_vpc_attachment" "spoke" {  
  provider      = aws.spoke  
  transit_gateway_id = var.tgw_id  
  vpc_id        = var.vpc_id  
  subnet_ids    = var.transit_subnet_ids  
}
```

```
resource "aws_ec2_transit_gateway_route_table_association" "spoke" {  
  provider      = aws.hub  
  transit_gateway_attachment_id = aws_ec2_transit_gateway_vpc_attachment.spoke.id  
  transit_gateway_route_table_id = var.spoke_route_table_id  
}
```

Context required: RAM share, accepted attachment and explicit routing controls.

Appendix B: Linux ♦ Windows groups

```
resource "aws_eks_node_group" "os" {
  for_each      = toset(["linux", "windows"])
  cluster_name  = var.cluster_name
  node_group_name = each.key
  node_role_arn = var.node_role_arn
  subnet_ids    = var.private_subnet_ids
  ami_type      = each.key == "linux" ? "AL2023_x86_64_STANDARD" : "WINDOWS_CORE_2022_x86_64"
  instance_types = [var.instance_type_by_os[each.key]]
  scaling_config {
    desired_size = 2
    min_size     = 2
    max_size     = 4
  }
}
```

Existing EKS cluster, IAM, Windows IPAM and node authorization required.

Appendix C: Linux diagnostic sample

```
#!/usr/bin/env bash
set -uo pipefail
service="nginx" # approved service name

timeout 10s systemctl is-active "$service" || true
timeout 10s journalctl -u "$service" --since "15 minutes ago" \
  --no-pager -n 100 || true
free -m
df -hP /
df -iP /
```

Read-only excerpt. Send bounded output to encrypted, access-controlled evidence storage.

Appendix D: Windows diagnostic sample

```
$ErrorActionPreference = "Continue"
Get-Service -Name W3SVC -ErrorAction SilentlyContinue
Get-Volume | Select-Object DriveLetter, SizeRemaining, Size
Get-WinEvent -FilterHashtable @{
    LogName = "Application"
    StartTime = (Get-Date).AddMinutes(-15)
} -MaxEvents 20 -ErrorAction SilentlyContinue |
    Select-Object TimeCreated, Id, LevelDisplayName, Message
if (Get-Module -ListAvailable WebAdministration) {
    Import-Module WebAdministration
    Get-ChildItem IIS:\AppPools | Select-Object Name, State
}
```

Read-only excerpt. SSM timeout and evidence redaction are set in the wrapper.